Is That Supernode Special?

Selection Rules and Null Models Change What Attribution-Graph Interventions Validate

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Abstract

Circuit-tracing studies support their claims by suppressing or swapping labelled groups of transcoder features (supernodes), but a successful intervention does not show that the *labelled* features are what matters. We test such claims against matched random replacements of their features, under an activation-matched and an attribution-influence-matched null model, using 2,958 automatically constructed claims and 14 hand-curated ones from two models, two transcoder types and two task families. Thirteen of the 14 hand-curated claims pass. With per-layer transcoders, only 0–56% of the claims built with a contrastive selection rule are specific; a held-out causal selection rule raises this share to 44–77% but does not help with cross-layer transcoders. On two-hop recall, the activation-matched null declares 88–99% of the contrastive per-layer claims specific, whereas on subject–verb agreement the two nulls agree. Most two-hop ablation claims that pass the stricter test beat their null by less than a factor of two. Because the selection rule and the null model can change the verdict on a claim, we give reporting recommendations and release the test as a library.

1 Introduction

Attribution graphs (Ameisen et al., 2025; Lindsey et al., 2025) and the open-source library *circuit-tracer* (Hanna et al., 2025) have made feature-level circuit analysis widely available. A typical analysis replaces the MLPs with transcoders, attributes the output to transcoder features, and groups important features into labelled *supernodes* such as "Texas". The analyst then tests the resulting account with interventions. For example, suppressing the "Texas" supernode $\times(-2)$ removes "Austin" from the output, and additionally injecting "California" features makes the model say "Sacramento".

These interventions carry the evidential weight, but they answer a weaker question than the one they are used for. They show that perturbing *these* features changes the output. They do not show that the effect is specific to these features rather than to any comparably important set of features at the same position. In a two-hop prompt, the last city token may host many active features that contribute to the answer, and ablating ten of them may remove the answer whichever ten are chosen.

Evaluations of sparse dictionaries and steering methods already compare against baselines, and circuit discovery has robustness checks (Section 2). Supernode interventions are usually reported without a null model, even as automated pipelines for annotating attribution graphs begin to rely on them as evidence (Birardi and Paulo, 2025; Arora et al., 2026; Patel et al., 2026).

We make this question testable (Figure 1). For each intervention claim, we repeatedly replace the labelled features with matched random features and compute an empirical p-value, using either an activation-matched or an attribution-influence-matched null model. We apply this *specificity test*, with multiple-comparison control, to 2,958 automatically constructed claims from two models, two transcoder types and two task families, and to 14 hand-curated claims with 5,000 null draws each, and release it as a library (Section 3). We find that (1) the rule used to select supernode features changes which claims pass, and the direction of this effect depends on the transcoder type (Section 5.2); (2) the choice of null model changes many verdicts

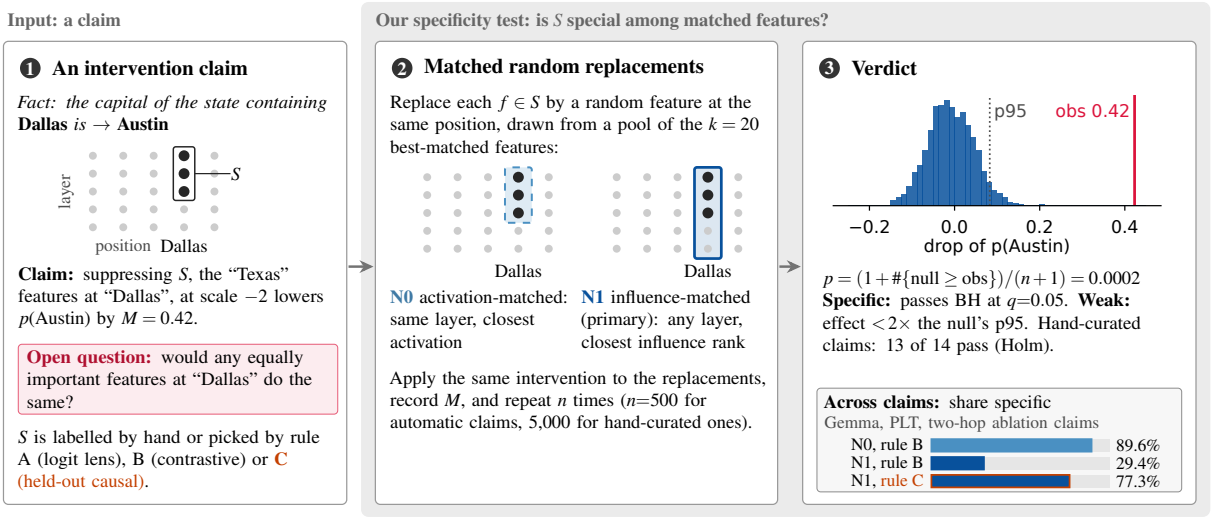


Figure 1: Testing an intervention claim against matched nulls. (1) A claim consists of a supernode S, an intervention and a claim-specific metric; here S is the "Texas" supernode at the last city token (claim D3), suppressed $\times(-2)$, with the drop in p(Austin) as the metric. S is labelled by hand or selected automatically by rule A, B or C (Section 3). (2) Each feature of S is replaced by a random feature at the same position. The grids (layers $\times$ positions, as in panel 1) show the candidate pools: N0 draws from the 20 features in the same layer with the closest activation, and N1 (our primary null) from the 20 features in any layer with the closest attribution-influence rank. The replacement set receives the same intervention, and this is repeated n times. (3) The observed effect is compared with the resulting null distribution to give an empirical p-value. Claims that pass Benjamini–Hochberg correction at $q = 0.05$ are called specific, and weak if their effect is below twice the null's 95th percentile. The histogram shows the N1 null of claim D3 ($n = 5{,}000$). The inset gives the share of specific claims for Gemma PLT two-hop ablation claims (Table 2 and Appendix B).

on two-hop recall but not on subject–verb agreement, and a probe tests one explanation for this difference (Section 5.3); (3) on two-hop recall, most ablation claims that pass the test beat the null by less than a factor of two (Section 5.4). We recorded all hypotheses, decision rules and sample sizes in a dated design log before running the corresponding experiments; the log was not externally registered (Appendix A).

2 Background and Related Work

Attribution graphs. A replacement model substitutes each MLP with a per-layer transcoder (PLT), or all MLPs with a single cross-layer transcoder (CLT). With attention patterns frozen, the residual stream can then be attributed linearly to features, error nodes and token embeddings (Dunefsky et al., 2024; Ameisen et al., 2025; Paulo et al., 2025). circuit-tracer implements interventions through `feature_intervention`, which sets chosen features to chosen values and propagates the effect with attention frozen. We use this function unchanged for every intervention in this paper. Case studies test supernodes by suppressing them or by swapping in the supernode of a donor prompt, as in the two-hop and multilingual examples of Lindsey et al. (2025). The planning case study from the same work has since been stress-tested on open models and open CLTs (Jacopin, 2026).

Baselines and robustness. Evaluations of dictionary features compare against random baselines (Heap et al., 2025; Korznikov et al., 2026), and steering methods are compared with simple baselines (Wu et al., 2025). For circuits, MIB compares methods for circuit localisation (Mueller et al., 2025), CIRCUS builds consensus circuits by ensembling over pruning thresholds (Parekh, 2026), and Bayat Makou et al. (2026)

show that structurally different circuits can implement the same computation. None of these provides a null distribution for the effect of a feature-level intervention claim, which is what we construct here.

PLT versus CLT. Lange et al. (2026) argue that CLT training can favour shallow rewrites of multi-hop circuits. We compare the two transcoder types only as a secondary factor, for the reason given in the limitations (Section 6).

3 Method

Claims. We write an intervention claim as a prompt x, a set of feature groups $G_1, \ldots, G_m$ and an output metric M. Each group is a set of (layer, position, feature) triples taken from a source prompt, which is either x or a donor prompt. The intervention sets every feature in group G_j to s_j times its activation on the source prompt, optionally at a different position of x. The metric M is computed on the output distribution at the final position and is oriented so that larger values mean the claim holds more strongly. We consider two claim types. An *ablation* claim suppresses the supernode S at scale -2 and measures the drop in $p(\text{answer})$. A *swap* claim also injects the donor's supernode S' at scale $+2$ at the same position and measures $p(\text{donor answer})$. The metric has to be fixed for each claim before testing. For example, the claim that suppressing French features makes the model answer in English looks non-specific under a generic metric, the drop in top-1 probability ($p = 0.08$), but is highly specific under the metric it actually claims, $p(\text{“big”})$ ($p = 0.002$), with the same 5,000 null draws.

Null models. For each labelled feature f, we draw a random replacement and apply the same intervention to it. All labelled features of the prompt and its donor are excluded from the candidate pools. We use two nulls. N0 (activation-matched) draws uniformly from the $k = 20$ active features at the same layer and position whose activations are closest to that of f. This corresponds to the familiar random-feature baseline. N1 (influence-matched), our primary null, ranks features by their attribution influence on the target logits in the source prompt's graph (at most 8,192 feature nodes) and draws from the $k = 20$ features at the same position with the closest rank. It asks whether the labelled features are special among features that the graph itself treats as equally important. For swap claims we also use N1 inject-only, which keeps the real ablation and randomises only the injected group. With n draws, we compute $p = (1 + \#\{\text{null} \geq \text{obs}\})/(n+1)$ (Phipson and Smyth, 2010).

Statistics. Within each family of claims (defined by task, model, transcoder, claim type, selection rule and null), we call a claim *specific* if it passes Benjamini–Hochberg correction at $q = 0.05$ (Benjamini and Hochberg, 1995), and we report proportions with Wilson intervals (Wilson, 1927). For the hand-curated claims we use Holm correction at $\alpha = 0.01$ (Holm, 1979). Over 14 claims, the smallest p must then be at most $0.01/14 \approx 7 \cdot 10^{-4}$, which requires at least about 1,400 draws; we use $n = 5{,}000$. We call a specific claim *weak* if its effect is below twice the 95th percentile of its null.

Selection rules. Automatic annotation needs a rule that picks the features representing an intermediate variable, at the position where that variable should be represented. We compare three rules. Rule A keeps the features whose logit-lens top-10 tokens include the first token of the intermediate entity. Rule B (contrastive, $K = 10$) takes the ten features with the largest positive $a_{\text{prompt}} - a_{\text{donor}}$. Rule C (held-out causal, $K = 10$) uses a *sibling* prompt that shares the intermediate variable. We zero-ablate each active feature on the sibling prompt, keep the ten with the largest drop in the sibling's own metric, and retain those that are also active on the test prompt. Because the selection never uses the output on the test prompt, it cannot favour features on the basis of the outcome being tested.

Tests and decision rules. The *null-leniency test* asks whether N0 declares more claims specific than N1, using an exact one-sided McNemar test on the same claims (McNemar, 1947). We consider this leniency to generalise across tasks if it holds in the same direction with $p < 0.01$ for both models on the second task. The *selection-rule test* asks whether rule C yields more specific claims than rule B under N1, using an exact

Table 1: Hand-curated claims under N1 ($n = 5{,}000$): observed claimed metric, null 95th percentile, p, and Holm verdict over all 14 claims ($\alpha = 0.01$). $\downarrow$: drop of; *tutorial: "little effect"; "ba" is the first token of "baht"; Scandinavian languages: Swedish+Danish+Norwegian. D4 is sensitive to the GPU type (Section 5.1).

Claim	Intervention $\rightarrow$ claimed metric	Obs	p95	p	Holm
D1	"say a capital" $\times -2 \rightarrow p$(Texas)	0.460	0.185	0.0002	pass
D2	"capital" $\times -2 \rightarrow \downarrow p$(Austin)	0.407	0.106	0.0002	pass
D3	"Texas" $\times -2 \rightarrow \downarrow p$(Austin)	0.423	0.083	0.0002	pass
D4	"state"* $\times -2 \rightarrow \downarrow p$(Austin)	0.051	0.039	0.016	fail
D5	Texas $\times -2$ + California $\times 2 \rightarrow p$(Sacramento)	0.364	0.005	0.0002	pass
D6	Texas $\times -2$ + China $\times 2 \rightarrow p$(Beijing)	0.212	<0.001	0.0002	pass
M1	French $\times -2 \rightarrow p$("big")	0.486	0.075	0.002	pass
M2	French $\times -2$ + Chinese $\times 2 \rightarrow p$(Chinese "big")	0.846	<0.001	0.0002	pass
M3	small $\times -2$ + big $\times 2 \rightarrow p$("petit")	0.863	0.012	0.0002	pass
L1	China $\times 0 \rightarrow \downarrow p$(yuan)	0.381	0.031	0.0002	pass
L2	China $\times 0$ + Thailand/SEA $\times 1 \rightarrow p$("ba")	0.638	0.015	0.0002	pass
L3	China $\times 0$ + Thailand $\times 2$ at colon $\rightarrow p$("ba")	0.275	0.002	0.0002	pass
L4	currency $\times 0$ + language $\times 5 \rightarrow p$(Chinese)	0.164	0.027	0.0002	pass
L5	US $\times 0$ + Scandinavia/Denmark $\times 5 \rightarrow p$(Scandinavian languages)	0.451	0.036	0.0002	pass

two-sided McNemar test. Its interpretation was fixed before rule C was run. If rule C reaches at least 70% specific claims and the test is significant, we attribute the low specificity under rule B mainly to the *selection rule* rather than to an absence of specific features. If rule C stays at or below 40%, we rule this explanation out. Otherwise, we conclude that *both* contribute.

4 Setup

Models. We study Gemma-2-2B (Gemma Team et al., 2024) and Llama-3.2-1B (Grattafiori et al., 2024). For Gemma we use the Gemma Scope per-layer transcoders (Lieberum et al., 2024), taking the lowest-L_0 transcoder at each layer, and a CLT with 426k features. For Llama we use per-layer transcoders and a CLT with 524k features. All of these are public checkpoints distributed with circuit-tracer, and all models run in bf16.

Two-hop recall. We use 200 prompts of the form "Fact: the capital of the {state|country} containing {city} is", covering 50 states and 50 countries with two cities each. Every prompt therefore has exactly one sibling, the other city in the same region. We keep the prompts that each model answers correctly, which leaves 163 for Gemma and 92 for Llama. Each prompt has a fixed donor from a different region.

Subject–verb agreement (SVA). We build 200 prompts from two templates without attractor nouns, "The {noun} that I saw yesterday" and "The {noun} that we talked about", using 50 nouns that are single tokens in both singular and plural. The metric is the share of the correct grammatical number among the verb continuations {is, was, has} and {are, were, have}. We keep prompts where this share exceeds 0.5 and the two verb sets together receive probability at least 0.1, which leaves 135 prompts for Gemma and 110 for Llama. The donor of each prompt is its number-flipped version. Rule A is not applicable here, because there is no natural label target at the subject position.

Hand-curated claims. We take nine Gemma claims from the circuit-tracer tutorial (two-hop D1–D6 and multilingual M1–M3) and five Llama claims from its Llama demo (L1–L5), copying the supernodes and intervention scales exactly. We test each claim with its claimed metric under N1 with $n = 5{,}000$ draws.

Sizes and compute. For the automatic claims we use $n = 500$ draws for N1 and for the inject-only null, and $n = 250$ for N0. With $n = 500$, the smallest attainable p is 0.002. The study used about 130 GPU-hours on A800 and H200 GPUs.

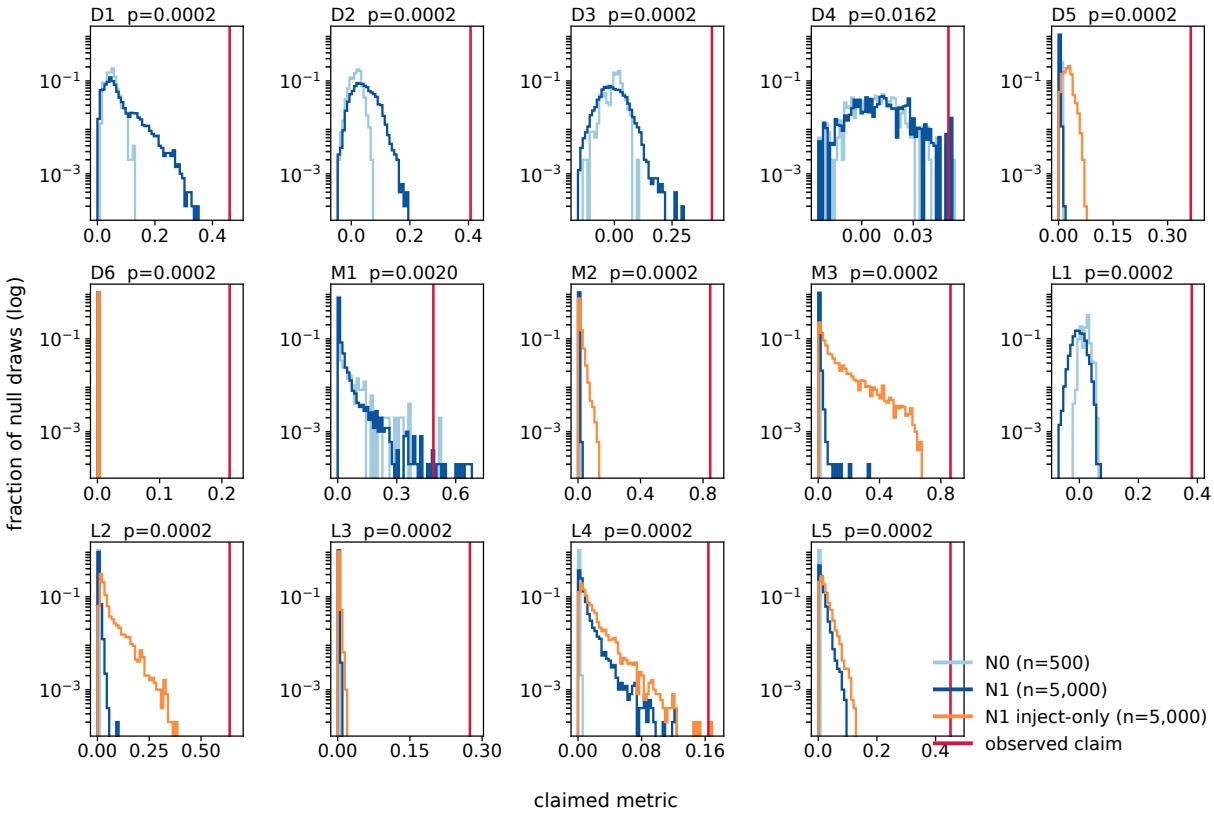


Figure 2: Null distributions of the 14 hand-curated claims: N0 ($n = 500$), N1 ($n = 5{,}000$) and, for swap claims, N1 inject-only ($n = 5{,}000$), with the observed claim in red. Panel titles give the claim (D, M: Gemma; L: Llama) and its N1 p-value (Table 1).

5 Results

5.1 Hand-Curated Claims Pass

Thirteen of the fourteen claims pass (Table 1), and every swap claim also passes its inject-only null ($p \leq 0.0004$), so it matters which donor features are injected. The one failure, D4, is a claim that the tutorial itself describes as having little effect; its effect is only 1.3 times the 95th percentile of its null. Passing the test does not mean that only the labelled features can produce the effect. For M1, some random suppressions exceed the observed effect (null maximum 0.67 against 0.49; Figure 2), and for M3 the inject-only null reaches 0.67 against an observed 0.86.

Sensitivity to hardware. We reran the 14 claims with the same code and seeds on a second GPU type (NVIDIA A800 instead of H200). Differences in bf16 arithmetic changed the observed effects by up to 0.027 and, through small changes in the attribution ranking, the N1 candidate pools. Twelve claims stayed at the minimum attainable p of 0.0002, and M1 moved to 0.0006. D4's effect fell from 0.051 to 0.042 and its p to 0.0064, so it passes on the second GPU type; its verdict is not robust to numerical details. The automatic claims were computed on both GPU types, so verdicts close to the threshold may change on other hardware as well.

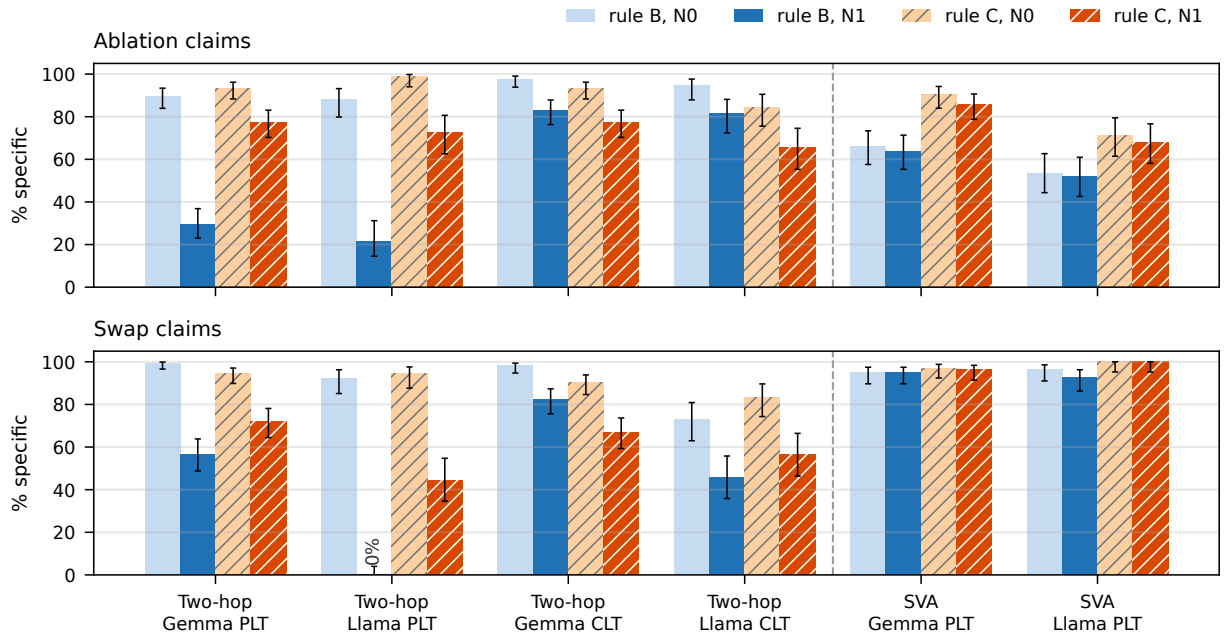


Figure 3: Share of claims judged specific (BH $q = 0.05$; Wilson 95% intervals) per setting, claim type, rule (B contrastive, C held-out causal) and null (N0 light, N1 dark). Left of the dashed line: two-hop recall; right: SVA.

5.2 The Selection Rule Changes Which Claims Pass

With rule B, 0–56% of PLT supernodes support their claims under N1 (Table 2, Figure 3; the activation-matched results are in Appendix B). In Llama's PLT swaps, no supernode supports its claim. The median swap effect (0.0072) is below the median null mean (0.018), so replacing both the suppressed and the injected supernode with equally important random features moves the output further toward the donor's capital than the selected pair does. When the real suppression is kept and only the injection is randomised, the median null mean drops to 0.0053; the gap therefore arises from randomising the suppressed set, not the injected one. Rule A selects features at the city position for few prompts (Gemma 11/163, Llama 24/92). On the 24 Llama prompts where both rules produce an ablation claim, rule A declares none specific and rule B six (agreement 75%, $\kappa = 0.00$).

For both PLT models, rule C yields many more specific claims than rule B. For ablations the share rises from 29% to 77% for Gemma and from 22% to 73% for Llama; for swaps it rises from 56% to 72% and from 0% to 44%. The disagreements between the two rules go almost entirely one way (84 claims specific only under C against 6 only under B for Gemma ablations, and 40 against 0 for Llama swaps). By our decision rule, the low shares under rule B are mainly due to the selection rule in three of the four PLT comparisons, and both factors contribute for Llama swaps. A transcoder-free check rules out another explanation for Llama's 0% swap rate. Patching the residual stream at the city position with the donor's activations raises $p(\text{donor capital})$ by a median of 0.13 (largest increase over four layers), so a swappable representation exists, and rule B fails to find it.

With CLTs the pattern differs. Contrastive CLT supernodes are specific much more often than PLT ones (46–83% against 0–56%), and rule C does not improve on them; it is significantly worse for Llama ablations (21 against 7 discordant claims) and for Gemma swaps (38 against 13). One untested explanation is that a CLT feature at the city position writes to all later layers, so that the contrastive difference at that position

Table 2: Two-hop recall under N1: share of specific claims with rule B (contrastive) and rule C (held-out causal). Discordant: claims specific only under C / only under B; p: exact two-sided McNemar. Verdict (ablation / swap) per the decision rule of Section 3: *rule*, the low rule-B share is mainly due to the selection rule; *both*, both factors contribute. Rule-C shares are over prompts with a non-empty rule-C set (Llama: 90–91 of 92; Appendix A).

	Ablation			Swap			
Model · transcoder (n)	B → C	C-only / B-only	p	B → C	C-only / B-only	p	Verdict
Gemma · PLT (163)	29.4% → **77.3%**	84 / 6	1e-18	56.4% → **71.8%**	42 / 17	0.0015	rule / rule
Llama · PLT (92)	21.7% → **72.5%**	51 / 5	1e-10	0.0% → **44.4%**	40 / 0	2e-12	rule / both
Gemma · CLT (163)	82.8% → 77.3%	16 / 25	0.21	82.2% → **66.9%**	13 / 38	0.0006	both / both
Llama · CLT (92)	81.5% → **65.6%**	7 / 21	0.013	45.6% → 56.7%	17 / 8	0.11	both / both

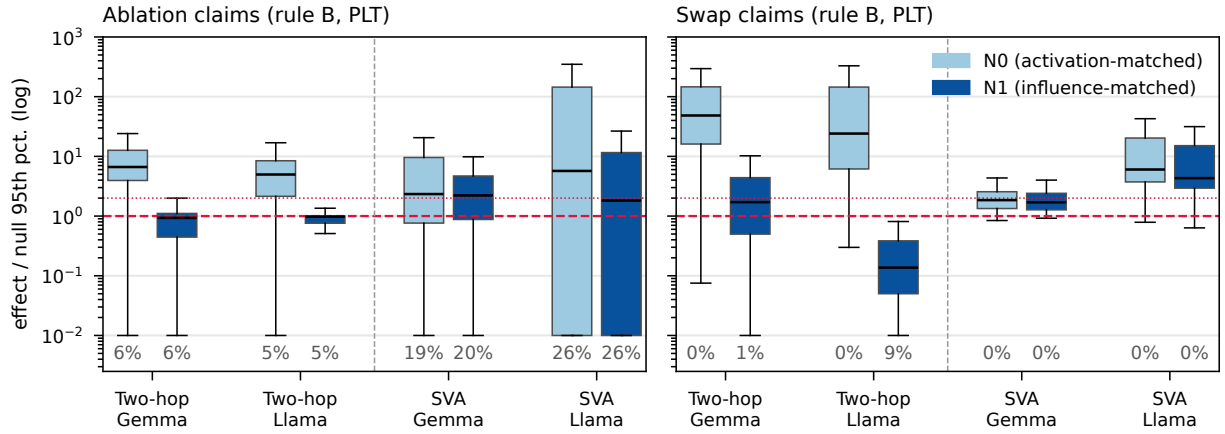


Figure 4: Effect divided by the null's 95th percentile (rule B, PLT) under N0 and N1. On two-hop recall the ratios under N0 lie well above those under N1, whereas on SVA the two distributions overlap heavily. Dashed: ratio 1; dotted: ratio 2 ("weak" threshold). Values below 0.01 (incl. negative) are clipped; grey numbers give the clipped share.

already captures the features that carry the variable forward, while single-feature ablation on a sibling prompt favours features whose effects are redundant.

5.3 The Null Model Changes Verdicts on Two-Hop Recall

On two-hop recall, the activation-matched null is much more lenient than N1. For every model, transcoder, rule and claim type, between 13 and 98 claims per family are specific only under N0, and at most two only under N1 (null-leniency test, all $p < 10^{-3}$; Table 3 and Table 6). On SVA the two nulls agree, so by our criterion the leniency of N0 does not generalise across tasks. The ratio of each claim's effect to its null's 95th percentile shows the same contrast (Figure 4). For PLT claims on two-hop recall (rules B and C), the median ratio is 2–61 under N0 but 0.1–5 under N1. On SVA the distributions overlap heavily; the medians are close for Gemma (2.5 against 2.3 for rule-B ablations) but not for Llama (5.7 against 1.8).

One explanation is that the city position hosts many active features that barely affect the answer, so that activation-matched replacements are nearly inert and N0 is an easy bar. We tested it with a probe. For 60 claims per task and model, we enumerated both null pools, intervened on each candidate alone, scored it with the claim's metric, and summarised each claim by the ratio R of the mean single-feature effect in the

Table 3: Share (%) of specific claims under N0 and N1 (PLT) and the one-sided p of the null-leniency test. Discordant counts for all families are in Table 6.

		Ablation			Swap		
	Rule	N0	N1	p	N0	N1	p
2-hop Gemma	B	90	29	3e-30	99	56	8e-22
2-hop Gemma	C	93	77	1.5e-8	94	72	7e-12
2-hop Llama	B	88	22	4e-19	92	0	3e-26
2-hop Llama	C	99	73	6e-8	94	44	3e-14
SVA Gemma	B	66	64	0.31	95	95	0.69
SVA Gemma	C	90	86	0.055	97	96	0.50
SVA Llama	B	54	52	0.31	96	93	0.063
SVA Llama	C	71	68	0.19	100	100	1

Table 4: Share (%) of specific claims (N1, BH) that are *weak*, meaning that their effect is below twice the null's 95th percentile; rule B / rule C.

Task · model · transcoder	Ablation B / C	Swap B / C
Two-hop · Gemma · PLT	90 / 74	23 / 9
Two-hop · Llama · PLT	95 / 95	– / 5
Two-hop · Gemma · CLT	91 / 76	10 / 8
Two-hop · Llama · CLT	97 / 90	12 / 6
SVA · Gemma · PLT	16 / 15	60 / 33
SVA · Llama · PLT	5 / 8	5 / 0

N0 pool to that in the N1 pool. R was lower on two-hop recall than on SVA for both models (median 0.12 against 0.68 for Gemma, one-sided Mann–Whitney $p = 3 \cdot 10^{-7}$; 0.06 against 0.34 for Llama, $p = 0.002$). At the city position, N0-pool features have a median single-feature effect near zero (10^{-4} for Gemma and $-5 \cdot 10^{-4}$ for Llama, against 0.004 in the N1 pool), and they rank much lower in the attribution graph (median influence percentile 0.43–0.50, against 0.19–0.27 for the supernode). The probe supports the explanation only for the aggregate difference between tasks: R is undefined for 22–40% of claims, a secondary summary shows similar gaps on both tasks, and R does not predict which individual claims are specific only under N0 (Appendix B).

5.4 Most Specific Two-Hop Ablation Claims Are Weak

On two-hop recall, most specific ablation claims beat N1 only narrowly. For every model, transcoder and rule, 74–97% of them are weak (Table 4), and the median ratio of effect to null 95th percentile across all ablation claims is 0.9–1.2. Suppressing ten equally important features at the city position removes a similar share of the answer. Swap claims are rarely weak (5–23%). On SVA, most specific ablation claims are strong, with only 5–16% weak.

The intervention scale also changes verdicts. On the first 40 Gemma PLT prompts with rule B, 20, 13 and 16 of the 40 ablation claims were specific at scales -1, -2 and -4 (Figure 5).

6 Discussion

What is validated. An intervention demonstration supports the claim that these features, perturbed in this way, change the output. A specificity test adds a second claim, that comparably important features do not.

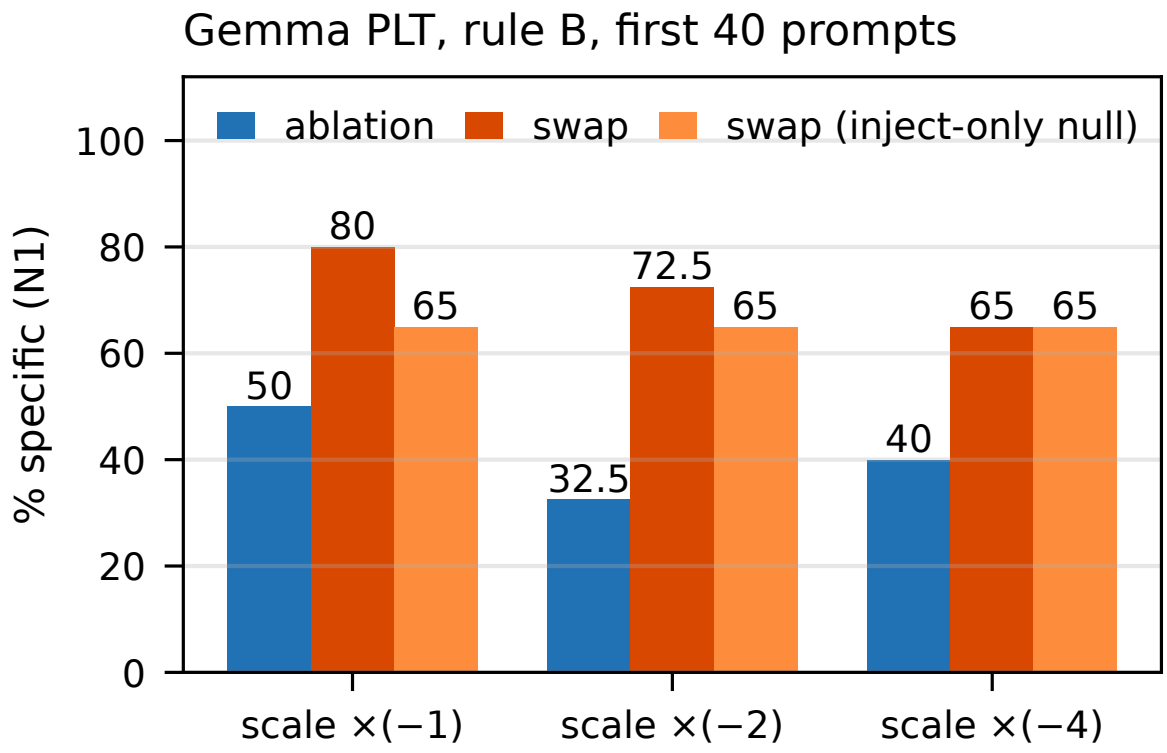


Figure 5: Share of specific claims (N1, BH within 40 claims) at suppression scales -1, -2, -4; Gemma PLT, rule B, the first 40 two-hop prompts in list order. Null draws: $n = 250$ at -1 and -4; the main runs ($n = 500$) at -2.

In our experiments the second claim held for the curated examples, which may have been chosen because they worked. It often failed for the contrastive heuristic with per-layer transcoders. Automatic annotation pipelines should therefore report their selection rule together with a matched-null test.

Which null. N1 matches the comparison an analyst implicitly makes, against other features that the graph deems important at the same place, and in no family was it more lenient than N0. Because the leniency of N0 varies by task (Section 5.3), N0 is informative only when reported alongside N1. N1 inherits the graph's notion of importance, so graph-free checks such as rule C and residual patching complement it.

Recommendations.

1. Fix a claim-specific metric in advance.
2. Report the influence-matched p-value with enough draws for the planned correction ($n \geq 2{,}000$ for Holm over about ten claims).
3. Report the effect-to-null ratio and the upper tail of the null, and describe claims close to the null as borderline.
4. Report the supernode-selection rule and compare it with at least one alternative, since the better rule depended on the transcoder.
5. Report sensitivity to the intervention scale.

6. Test on more than one task family before generalising.

Limitations. We study two small models, two task families and one intervention protocol, with attention frozen. The null design fixes $k = 20$ and the attribution budget, and we did not vary them. Rule C requires a sibling prompt that shares the intermediate variable, which is not always available. The PLT and CLT dictionaries differ in training data, width and sparsity, so the PLT–CLT differences are not causal effects of the transcoder type. The scale sensitivity was measured on the first 40 Gemma prompts in list order, not on a random subset. Borderline verdicts depend on numerical details such as the GPU type (Section 5.1).

7 Conclusion

Whether an intervention claim from an attribution graph passed a matched-null test depended on how its features were selected, on the transcoder type and, on two-hop recall, on the null model. We suggest that such tests be reported alongside intervention results, together with the selection rule and the effect-to-null ratio. In our view, an intervention is evidence about the labelled features only when it clears an influence-matched null by a clear margin; a pass by less than a factor of two, as for most automatically selected two-hop supernodes, shows little more than that the position matters.

Software and Data

We provide the code for the test (with unit tests), the prompt sets, the experiment and analysis scripts, the dated design log and the records of all runs as anonymous supplementary material, and will release them.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning, specifically the reliability of interpretability evidence. Better-calibrated claims about model internals may support safer model auditing. We do not foresee specific negative societal consequences.

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A Design Log and Deviations

Each row lists the decisions recorded in the design log before the corresponding runs.

Before	Recorded decisions
Hand-curated claims	Null models ($k = 20$; N1 primary), the claimed metric of each claim, $n = 5{,}000$ draws and Holm at $\alpha = 0.01$ over all 14 claims.
Automatic claims	Rules A and B; BH within families; the null-leniency test, the agreement of rules A and B, and the weak criterion; $n = 500$ for N1 and the inject-only null, $n = 250$ for N0.
Second model	Replication on Llama-3.2-1B with falsification conditions.
Rule C	Rule C, the selection-rule test and its decision rule; the residual-patching check.
Second task	SVA prompts, metric and correctness criterion; rules B and C; the cross-task criterion for the null-leniency test.
Mechanism probe	Probe design and the decision rule on R (outcome: supported); the claim-level analysis as report-only.

The 2,958 automatic claims in the main text are the rule-B and rule-C claims. Rule A yields 39 claims, because it selects features for few prompts, and the scale-sensitivity runs add 160 claims at scales -1 and -4.

Deviations. (i) CLT arms were first run on the first 40 prompts (in list order) and later extended; the first-40 estimates under rule B were 2–18 points lower than the full-sample ones. (ii) Rule C's selected set is empty for some prompts, which then have no rule-C ablation claim (Gemma SVA 2/135; Llama two-hop 1/92 with PLT and 2/92 with CLT; Llama SVA 16/110). A rule-C swap claim also needs a non-empty set on the donor side, which leaves 131/135 (Gemma SVA), 90/92 (Llama two-hop, PLT and CLT) and 77/110 (Llama SVA) swap claims. Rule-C shares use these counts as denominators. (iii) One SVA prompt lies at the probability-mass threshold (0.102) and was classified as correct on one GPU type and as incorrect on the other; all runs use the first classification (correct).

B Additional Tables

Table 5: Two-hop recall, rule B, all transcoders: share specific under N0 and N1 (Wilson 95% CI).

Model · transcoder (n prompts)	Ablation N0	Ablation N1	Swap N0	Swap N1
Gemma · PLT (163)	89.6%	29.4% [23.0, 36.9]	99.4%	56.4% [48.8, 63.8]
Llama · PLT (92)	88.0%	21.7% [14.5, 31.2]	92.4%	0.0% [0, 4.0]
Gemma · CLT (163)	97.5%	82.8% [76.3, 87.8]	98.2%	82.2% [75.6, 87.3]
Llama · CLT (92)	94.6%	81.5% [72.4, 88.1]	72.8%	45.6% [35.9, 55.8]

Claim-level association between R and "specific only under N0" on two-hop recall (report-only): point-biserial $r = -0.18$ ($p = 0.21$) for Gemma and $r = +0.31$ ($p = 0.045$, opposite sign) for Llama. Secondary summary (share of pool features exceeding half the supernode's median effect), N0 / N1: two-hop 0.31/0.45 (Gemma), 0.32/0.40 (Llama); SVA 0.17/0.23, 0.27/0.37. Transcoder-free residual patching for Llama two-hop recall (per prompt, the largest increase of $p(\text{donor capital})$ over layers 2/5/8/11): median 0.13, at least 0.05 in 80% of prompts.

C Tool Interface

```
from explai.specificity import Claim, Group, SpecificityTester
```

Table 6: Null-leniency test for all families: share (%) specific under N0 and N1, claims specific only under N0 / only under N1, and exact one-sided McNemar p.

		Ablation					Swap				
Setting	Rule	N0	N1	only N0	only N1	p	N0	N1	only N0	only N1	p
2-hop Gemma PLT	B	89.6	29.4	98	0	$3.2{\cdot}10^{-30}$	99.4	56.4	70	0	$8.5{\cdot}10^{-22}$
2-hop Gemma PLT	C	93.2	77.3	26	0	$1.5{\cdot}10^{-8}$	94.5	71.8	37	0	$7.3{\cdot}10^{-12}$
2-hop Llama PLT	B	88.0	21.7	61	0	$4.3{\cdot}10^{-19}$	92.4	0.0	85	0	$2.6{\cdot}10^{-26}$
2-hop Llama PLT	C	98.9	72.5	24	0	$6{\cdot}10^{-8}$	94.4	44.4	45	0	$2.8{\cdot}10^{-14}$
2-hop Gemma CLT	B	97.5	82.8	24	0	$6{\cdot}10^{-8}$	98.2	82.2	26	0	$1.5{\cdot}10^{-8}$
2-hop Gemma CLT	C	93.2	77.3	26	0	$1.5{\cdot}10^{-8}$	90.2	66.9	38	0	$3.6{\cdot}10^{-12}$
2-hop Llama CLT	B	94.6	81.5	13	1	$9.2{\cdot}10^{-4}$	72.8	45.6	25	0	$3{\cdot}10^{-8}$
2-hop Llama CLT	C	84.4	65.6	19	2	$1.1{\cdot}10^{-4}$	83.3	56.7	24	0	$6{\cdot}10^{-8}$
SVA Gemma PLT	B	65.9	63.7	10	7	0.31	94.8	94.8	2	2	0.69
SVA Gemma PLT	C	90.2	85.7	8	2	0.055	97.0	96.2	2	1	0.50
SVA Llama PLT	B	53.6	51.8	3	1	0.31	96.4	92.7	4	0	0.063
SVA Llama PLT	C	71.3	68.1	4	1	0.19	100.0	100.0	0	0	1

Table 7: Mechanism probe (60 rule-B PLT claims per row). R: mean single-feature effect in the N0 pool over that in the N1 pool (undefined when the latter is non-positive). Effects: medians over claims of the pool-mean single-feature effect; ranks: median influence percentiles (0 = most influential).

Task · model	R defined / undefined	R median [IQR]	Effect S / N0 / N1	Rank S / N0 / N1
Two-hop · Gemma	47 / 13	0.12 [−0.04, 0.39]	0.022 / 0.0001 / 0.0041	0.19 / 0.50 / 0.35
Two-hop · Llama	43 / 17	0.06 [−0.29, 0.19]	0.022 / −0.0005 / 0.0036	0.27 / 0.43 / 0.34
SVA · Gemma	36 / 24	0.68 [0.40, 1.04]	0.0004 / 5e-6 / 2e-5	0.69 / 0.87 / 0.66
SVA · Llama	42 / 18	0.34 [0.01, 0.60]	0.0018 / 5e-6 / 6e-5	0.64 / 0.72 / 0.56

```
t = SpecificityTester(model)            # circuit-tracer ReplacementModel
claim = Claim(prompt, [Group(S, prompt, -2.0, None)],
              metric=lambda p: base - p[answer_id].item(), exclude=set(S))
t.test(claim, "influence", n=500, k=20, gen=torch.Generator().manual_seed(0))
# -> {"obs", "null_mean", "null_p95", "null_max", "p", "n"}
```